

**Response to Request for Information on the Development of an Artificial Intelligence (AI)
Action Plan, National Science Foundation NSF_FRDOC_0001-3479, dated 02/06/2025**

Title: Multi-domain Collaboration Supporting Evolution of AI-Energy Nexus

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Recommendation: establish a public/private collaboration forum to enable advancement of AI capabilities in conjunction with energy system transformation. The goal would be to inform policy adjustments (and other actions) that accommodate rapid capacity growth, technology integration, efficiency and resilience. In particular, new processes and policies will be needed support diversification and decentralization of energy system technologies and ownership; flexible transactional mechanisms involving new market dynamics; and regulatory processes that provide for diverse input and informed consent.

Rationale: the rapid growth of Artificial Intelligence sophistication and use corresponds closely and interactively with energy transformation. In particular, projected electrical power demand growth associated with AI computing needs will substantially accelerate trends in energy growth and electrification. At the same time, energy system technologies and operational processes are evolving rapidly – from centralized energy networks dominated by fossil fueled sources to systems characterized by greater diversity, flexibility, decentralization, and intelligence. Moreover, new energy systems (as well as other processes) may achieve substantial efficiency and resilience advantages through increasing levels of intelligent control.

Illustration: the electrical power system is experiencing a shift in fundamental behaviors as traditionally dominant synchronous electromechanical equipment (e.g., generators and motors) are displaced by power electronic (digital) technologies, such as power inverters and variable speed motor drives associated with renewable generation, heating and cooling, and large industrial processes. By virtue of increasingly powerful computing capabilities, these digital power systems provide agile dynamic behaviors – though they lack inherently stabilizing physics principles of legacy technologies. This digitization phenomenon already has impacted grid stability in areas with relatively high concentrations of inverter-based energy sources (e.g., wind and solar), especially in the absence of local energy storage to support stabilizing inputs. The anticipated proliferation of computational power loads (data centers) with characteristically dynamic demand and digital power converters will increase and accelerate this shift in dynamic grid characteristics. Meanwhile, an influx of large (stable) generating systems and variations in power distribution, storage, and control could provide requisite stabilizing influence if design and operational concepts advance rapidly. Successful system transformation hinges upon examination of alternative operational scenarios, technical analysis, and development of enabling business models; all likely to require policy adjustments.

Key elements: the proposed collaboration structure should include public/private membership, resources, processes, and communication pathways to provide for:

- trending and analysis of market processes and energy/operational technology interactions and respective impacts to resilience (security, stability, sustainability, and welfare);
- multi-domain (technology, business, economic, social) analysis to support evolution of technology standards, business rules, and policy/regulatory measures;
- communications to disseminate insights and stakeholder interaction to maximize value and achieve informed consent.

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